

Blockchain Experiment 11

AIM: To implement a Hyperledger Fabric network, deploy chaincode, and perform asset creation, querying, and transfer operations.

PROCEDURE

Step 1: Update the system

Go to <https://shell.cloud.google.com> and run the following commands:

```
sudo apt update
sudo apt upgrade
-y
```

```
g2022_aditi_taksale@cloudshell:~$ sudo apt update
Get:1 https://cli.github.com/packages stable InRelease [3,917 B]
Get:2 https://packages.cloud.google.com/apt gcsfuse-noble InRelease [1,227 B]
Get:3 https://packages.cloud.google.com/apt cloud-sdk InRelease [1,620 B]
Get:4 https://apt.postgresql.org/pub/repos/apt noble-pgdg InRelease [180 kB]
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease
Get:6 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:7 https://packages.cloud.google.com/apt gcsfuse-noble/main amd64 Packages [66.1 kB]
Get:8 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0 InRelease [3,005 B]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:10 https://packages.cloud.google.com/apt cloud-sdk/main all Packages [1,982 kB]
Get:11 https://apt.postgresql.org/pub/repos/apt noble-pgdg/main amd64 Packages [675 kB]
Get:12 https://packages.cloud.google.com/apt cloud-sdk/main amd64 Packages [4,640 kB]
Hit:13 https://ppa.launchpadcontent.net/dotnet/backports/ubuntu noble InRelease
Get:14 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0/multiverse amd64 Packages [68.7 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [3,505 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2,377 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3,692 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,989 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [2,155 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,508 kB]
```

```
g2022_aditi_taksale@cloudshell:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
  apparmor libapparmor1
The following packages will be upgraded:
  docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin google-cloud-cli
  google-cloud-cli-anthoscli google-cloud-cli-app-engine-go google-cloud-cli-app-engine-java google-cloud-cli-app-engine-python
  google-cloud-cli-app-engine-python-extras google-cloud-cli-bigtable-emulator google-cloud-cli-cbt google-cloud-cli-cloud-run-proxy
  google-cloud-cli-datastore-emulator google-cloud-cli-gke-gcloud-auth-plugin google-cloud-cli-istioctl google-cloud-cli-kpt
  google-cloud-cli-local-extract google-cloud-cli-minikube google-cloud-cli-nomos google-cloud-cli-package-go-module
  google-cloud-cli-pubsub-emulator google-cloud-cli-scaffold google-cloud-cli-spanner-cli kubectrl libarchive13t64 libgdk-pixbuf-2.0-0
  libgdk-pixbuf2.0-common libpam-systemd libpq-dev libpq5 libruby3.2 libssl-dev libssl3t64 libsystemd-shared libsystemd0 libudev1
  linux-libc-dev openssl php8.3-bcmath php8.3-cgi php8.3-cli php8.3-common php8.3-dev php8.3-mbstring php8.3-mysql php8.3-opcache
  php8.3-readline php8.3-xml qemu-user qemu-user-static ruby3.2 ruby3.2-dev systemd systemd-dev systemd-sysv tzdata udev
59 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
Need to get 894 MB of archives.
```

Step 2: Install Required Dependencies

```
g2022_aditi_taksale@cloudshell:~$ sudo apt install -y jq curl git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
jq is already the newest version (1.7.1-3ubuntu0.24.04.1).
curl is already the newest version (8.5.0-2ubuntu10.8).
git is already the newest version (1:2.43.0-1ubuntu7.3).
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
```

Then verify everything is present:

```
node -v
```

```
npm -v
```

```
git --version
```

```
docker --version
```

```
docker ps
```

```
g2022_aditi_taksale@cloudshell:~$ node -v
v24.14.1
g2022_aditi_taksale@cloudshell:~$ npm -v
11.12.1
g2022_aditi_taksale@cloudshell:~$ git --version
git version 2.43.0
g2022_aditi_taksale@cloudshell:~$ docker --version
Docker version 29.4.0, build 9d7ad9f
g2022_aditi_taksale@cloudshell:~$
g2022_aditi_taksale@cloudshell:~$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
g2022_aditi_taksale@cloudshell:~$
```

Step 3: Fix Docker Compose

```
DOCKER_CONFIG=${DOCKER_CONFIG:-$HOME/.docker}
mkdir -p $DOCKER_CONFIG/cli-plugins
curl -SL
https://github.com/docker/compose/releases/download/v2.24.5/docker-compo
se
-linux-x86_64 \
  -o $DOCKER_CONFIG/cli-plugins/docker-compose
chmod +x
$DOCKER_CONFIG/cli-plugins/docker-compose docker
compose version
sudo mv /usr/bin/docker-compose /usr/bin/docker-compose-old 2>/dev/null ||
true
sudo tee /usr/local/bin/docker-compose > /dev/null <<'EOF'
#!/bin/bash
exec docker compose "$@"
EOF
sudo chmod +x /usr/local/bin/docker-compose
docker compose version
```

docker-compose version

```
g2022_anuprita_mhapankar@cloudshell:~$ DOCKER_CONFIG=${DOCKER_CONFIG:-$HOME/.docker}
mkdir -p $DOCKER_CONFIG/cli-plugins
curl -SL https://github.com/docker/compose/releases/download/v2.24.5/docker-compose-linux-x86_64 \
-o $DOCKER_CONFIG/cli-plugins/docker-compose
chmod +x $DOCKER_CONFIG/cli-plugins/docker-compose
docker compose version
% Total      % Received % Xferd Average Speed   Time    Time     Time  Current
             0      0     0             0             0             0             0
100 58.5M 100 58.5M    0     0  79.4M      0 --:--:-- --:--:-- --:--:-- 79.4M
Docker Compose version v2.24.5
g2022_anuprita_mhapankar@cloudshell:~$ sudo mv /usr/bin/docker-compose /usr/bin/docker-compose-old 2>/dev/null || true
g2022_anuprita_mhapankar@cloudshell:~$ sudo tee /usr/local/bin/docker-compose > /dev/null <<'EOF'
#!/bin/bash
exec docker compose "$@"
EOF
sudo chmod +x /usr/local/bin/docker-compose
g2022_anuprita_mhapankar@cloudshell:~$ docker compose version
docker-compose version
Docker Compose version v2.24.5
Docker Compose version v2.24.5
g2022_anuprita_mhapankar@cloudshell:~$
```

Step 4: Clone Fabric Samples

```
cd $HOME
git clone https://github.com/hyperledger/fabric-samples
cd fabric-samples
```

```
g2022_anuprita_mhapankar@cloudshell:~$ cd $HOME
g2022_anuprita_mhapankar@cloudshell:~$ git clone https://github.com/hyperledger/fabric-samples
Cloning into 'fabric-samples'...
remote: Enumerating objects: 15618, done.
remote: Counting objects: 100% (238/238), done.
remote: Compressing objects: 100% (129/129), done.
remote: Total 15618 (delta 161), reused 109 (delta 109), pack-reused 15380 (from 3)
Receiving objects: 100% (15618/15618), 24.07 MiB | 17.13 MiB/s, done.
Resolving deltas: 100% (8809/8809), done.
g2022_anuprita_mhapankar@cloudshell:~$ cd fabric-samples
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$
```

Step 5: Install Fabric Binaries and Docker Images

```
curl -sSLO
https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-
fabric.sh
chmod +x install-fabric.sh
./install-fabric.sh docker samples binary
```



```
g2022 anuprita_mhapankar@cloudshell:~/fabric-samples$ curl -sSLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
g2022 anuprita_mhapankar@cloudshell:~/fabric-samples$ chmod +x install-fabric.sh
./install-fabric.sh docker samples binary
```

Clone hyperledger/fabric-samples repo

==> Already in fabric-samples repo
fabric-samples v2.5.15 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.

Pull Hyperledger Fabric binaries

```
==> Downloading version 2.5.15 platform specific fabric binaries
==> Downloading: https://github.com/hyperledger/fabric/releases/download/v2.5.15/hyperledger-fabric-linux-amd64-2.5.15.tar.gz
==> Will unpack to: /home/g2022 anuprita_mhapankar/fabric-samples
% Total    % Received % Xferd Average Speed   Time    Time     Time  Current
           Dload Upload   Total   Spent    Left   Speed
  0     0    0     0    0     0      0      0  --:--:-- --:--:-- --:--:--    0
100 120M  100 120M    0    0 22.5M    0  0:00:05  0:00:05 --:--:-- 24.4M
==> Done.
```

```
==> Downloading version 1.5.17 platform specific fabric-ca-client binary
==> Downloading: https://github.com/hyperledger/fabric-ca/releases/download/v1.5.17/hyperledger-fabric-ca-linux-amd64-1.5.17.tar.gz
==> Will unpack to: /home/g2022 anuprita_mhapankar/fabric-samples
% Total    % Received % Xferd Average Speed   Time    Time     Time  Current
           Dload Upload   Total   Spent    Left   Speed
  0     0    0     0    0     0      0      0  --:--:-- --:--:-- --:--:--    0
100 31.6M  100 31.6M    0    0 15.9M    0  0:00:01  0:00:01 --:--:-- 31.6M
==> Done.
```

Pull Hyperledger Fabric docker images

```
FABRIC_IMAGES: peer orderer ccenv baseos
==> Pulling fabric Images
==> ghcr.io/hyperledger/fabric-peer:2.5.15
```

```
====> ghcr.io/hyperledger/fabric-ca:1.5.17
1.5.17: Pulling from hyperledger/fabric-ca
18afb8c8d2a4: Pull complete
6f4ebca3e823: Pull complete
b9649d3f054a: Pull complete
0b966e606d9a: Pull complete
c36bacd85367: Download complete
Digest: sha256:453a0a727e82c4960b797e379adc231e853ee23ae7526db53afef77b5074117e
Status: Downloaded newer image for ghcr.io/hyperledger/fabric-ca:1.5.17
ghcr.io/hyperledger/fabric-ca:1.5.17
==> List out hyperledger images
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
ghcr.io/hyperledger/fabric-baseos:2.5.15    654bd4554bf9    250MB    80.4MB
ghcr.io/hyperledger/fabric-ca:1.5.17       453a0a727e82    381MB    123MB
ghcr.io/hyperledger/fabric-ccenv:2.5.15    86dded7eda92    1GB      255MB
ghcr.io/hyperledger/fabric-orderer:2.5.15  9972c209be47    178MB    49.5MB
ghcr.io/hyperledger/fabric-peer:2.5.15     3e25adc31a75    230MB    66.2MB
hyperledger/fabric-baseos:2.5              654bd4554bf9    250MB    80.4MB
hyperledger/fabric-baseos:2.5.15           654bd4554bf9    250MB    80.4MB
hyperledger/fabric-baseos:latest            654bd4554bf9    250MB    80.4MB
hyperledger/fabric-ca:1.5                  453a0a727e82    381MB    123MB
hyperledger/fabric-ca:1.5.17                453a0a727e82    381MB    123MB
hyperledger/fabric-ca:latest                453a0a727e82    381MB    123MB
hyperledger/fabric-ccenv:2.5                86dded7eda92    1GB      255MB
hyperledger/fabric-ccenv:2.5.15             86dded7eda92    1GB      255MB
hyperledger/fabric-ccenv:latest             86dded7eda92    1GB      255MB
hyperledger/fabric-orderer:2.5              9972c209be47    178MB    49.5MB
hyperledger/fabric-orderer:2.5.15           9972c209be47    178MB    49.5MB
hyperledger/fabric-orderer:latest            9972c209be47    178MB    49.5MB
hyperledger/fabric-peer:2.5                 3e25adc31a75    230MB    66.2MB
hyperledger/fabric-peer:2.5.15              3e25adc31a75    230MB    66.2MB
hyperledger/fabric-peer:latest              3e25adc31a75    230MB    66.2MB
g2022 anuprita_mhapankar@cloudshell:~/fabric-samples$
```

Step 6: Verify Docker images

docker images | grep fabric

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ docker images | grep fabric
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
ghcr.io/hyperledger/fabric-baseos:2.5.15      654bd4554bf9      250MB      80.4MB
ghcr.io/hyperledger/fabric-ca:1.5.17          453a0a727e82      381MB      123MB
ghcr.io/hyperledger/fabric-ccenv:2.5.15       86dded7eda92      1GB        255MB
ghcr.io/hyperledger/fabric-orderer:2.5.15     9972c209be47      178MB      49.5MB
ghcr.io/hyperledger/fabric-peer:2.5.15        3e25adc31a75      230MB      66.2MB
hyperledger/fabric-baseos:2.5                 654bd4554bf9      250MB      80.4MB
hyperledger/fabric-baseos:2.5.15              654bd4554bf9      250MB      80.4MB
hyperledger/fabric-baseos:latest              654bd4554bf9      250MB      80.4MB
hyperledger/fabric-ca:1.5                     453a0a727e82      381MB      123MB
hyperledger/fabric-ca:1.5.17                  453a0a727e82      381MB      123MB
hyperledger/fabric-ca:latest                   453a0a727e82      381MB      123MB
hyperledger/fabric-ccenv:2.5                   86dded7eda92      1GB        255MB
hyperledger/fabric-ccenv:2.5.15                86dded7eda92      1GB        255MB
hyperledger/fabric-ccenv:latest                86dded7eda92      1GB        255MB
hyperledger/fabric-orderer:2.5                 9972c209be47      178MB      49.5MB
hyperledger/fabric-orderer:2.5.15              9972c209be47      178MB      49.5MB
hyperledger/fabric-orderer:latest              9972c209be47      178MB      49.5MB
hyperledger/fabric-peer:2.5                    3e25adc31a75      230MB      66.2MB
hyperledger/fabric-peer:2.5.15                 3e25adc31a75      230MB      66.2MB
hyperledger/fabric-peer:latest                 3e25adc31a75      230MB      66.2MB
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$
```

Step 7: Set Environment Variables

```
export PATH=$PATH:$HOME/fabric-samples/bin
```

```
export
```

```
FABRIC_CFG_PATH=$HOME/fabric-samples/config peer
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ export PATH=$PATH:$HOME/fabric-samples/bin
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ export FABRIC_CFG_PATH=$HOME/fabric-samples/config
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ peer version
peer:
  Version: v2.5.15
  Commit SHA: 83c7930
  Go version: go1.26.0
  OS/Arch: linux/amd64
  Chaincode:
    Base Docker Label: org.hyperledger.fabric
    Docker Namespace: hyperledger
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$
```

Step 8: Start the Network and Create Channel

```
cd $HOME/fabric-samples/test-network
```

```
./network.sh up createChannel
```

```

g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ export PATH=$PATH:$HOME/fabric-samples/bin
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ export FABRIC_CFG_PATH=$HOME/fabric-samples/config
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ ./network.sh up createChannel
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb with crypto from 'cryptogen'
Bringing up network
LOCAL_VERSION=v2.5.15
DOCKER_IMAGE_VERSION=v2.5.15
/home/g2022_anuprita_mhapankar/fabric-samples/test-network/./bin/cryptogen
Generating certificates using cryptogen tool
Creating Org1 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org1.yaml --output=organizations
org1.example.com
+ res=0
Creating Org2 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org2.yaml --output=organizations
org2.example.com
+ res=0
Creating Orderer Org Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-orderer.yaml --output=organizations
+ res=0
Generating CCP files for Org1 and Org2
[+] Running 3/7
! Network fabric_test Created
! Volume "compose_peer0.org1.example.com" Created
! Volume "compose_peer0.org2.example.com" Created
! Volume "compose_orderer.example.com" Created
! Container orderer.example.com Started
! Container peer0.org1.example.com Started
! Container peer0.org2.example.com Started
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
NAMES

```

```

/home/g2022_anuprita_mhapankar/fabric-samples/test-network/./bin/configtxgen
+ '[' 0 -eq 1 ']'
+ configtxgen -profile ChannelUsingRaft -outputBlock ./channel-artifacts/mychannel.block -channelID mychannel
2026-04-09 14:56:55.392 UTC 0001 INFO [common.tools.configtxgen.main] -> Loading configuration
2026-04-09 14:56:55.402 UTC 0002 INFO [common.tools.configtxgen.localconfig] completeInitialization -> orderer type: etcdraft
2026-04-09 14:56:55.402 UTC 0003 INFO [common.tools.configtxgen.localconfig] completeInitialization -> Orderer.EtcdRaft.Options unset, setti
n tick:10 heartbeat tick:1 max inflight blocks:5 snapshot interval size:16777216
2026-04-09 14:56:55.402 UTC 0004 INFO [common.tools.configtxgen.localconfig] Load -> Loaded configuration: /home/g2022_anuprita_mhapankar/f
configtx.yaml
2026-04-09 14:56:55.408 UTC 0005 INFO [common.tools.configtxgen] doOutputBlock -> Generating genesis block
2026-04-09 14:56:55.408 UTC 0006 INFO [common.tools.configtxgen] doOutputBlock -> Creating application channel genesis block
2026-04-09 14:56:55.410 UTC 0007 INFO [common.tools.configtxgen] doOutputBlock -> Writing genesis block
+ res=0
Creating channel mychannel
Adding orderers
+ . scripts/orderer.sh mychannel
+ '[' 0 -eq 1 ']'
+ res=0
Status: 201
{
  "name": "mychannel",
  "url": "/participation/v1/channels/mychannel",
  "consensusRelation": "consenter",
  "status": "active",
  "height": 1
}
Channel 'mychannel' created
Joining org1 peer to the channel...
Using organization 1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=0

```

```

Generating anchor peer update transaction for Org2 on channel mychannel
++ jq -r 'channel_group.groups.Application.groups.Org2MSP.values += {"AnchorPeers":{"mod_policy": "Admins","value":{"anchor_peers": [{"host": "peer0.org2.example.com","port": 90
51}],"version": "0"}})' /home/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/Org2MSPconfig.json
++ res=0
++ configtxlator proto_encode --input /home/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/Org2MSPconfig.json --type common.Config --output /home/g2022
_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/original_config.pb
++ configtxlator proto_encode --input /home/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/Org2MSPmodified_config.json --type common.Config --output /h
ome/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/modified_config.pb
++ configtxlator compute_update --channel_id mychannel --original /home/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/original_config.pb --updated /ho
me/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/modified_config.pb --output /home/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifa
cts/config_update.pb
++ configtxlator proto_decode --input /home/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/config_update.pb --type common.ConfigUpdate --output /home/g
2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/config_update.json
++ jq
+++ cat /home/g2022_anuprita_mhapankar/fabric-samples/test-network/channel-artifacts/config_update.json
++ echo '{"payload":{"header":{"channel_header":{"channel_id":"mychannel","type":2},"data":{"config_update":{"channel_id":"mychannel","isolated_data":{"{}","read
_set":{"groups":{"Application":{"groups":{"Org2MSP":{"groups":{"{}","mod_policy":{"","policies":{"Admins":{"mod_policy":{"","
","policy":{"null,"version":{"0"},"Endorsement":{"mod_policy":{"","policy":{"null,"version":{"0"},"Readers":{"mod_policy":{"","poli
cy":{"null,"version":{"0"},"Writers":{"mod_policy":{"","value":{"","values":{"Org2MSP":{"values":{"{}","mod_policy":{"","v
alue":{"null,"version":{"0"},"values":{"{}","version":{"0"},"values":{"{}","mod_policy":{"","
","policies":{"{}","values":{"{}","version":{"0"},"write_set":{"groups":{"Application":{"groups":{"Org2MSP":{"groups":{"{}","
mod_policy":{"Admins":{"mod_policy":{"","policy":{"null,"version":{"0"},"Endorsement":{"mod_policy":{"","poli
y":{"null,"version":{"0"},"Readers":{"mod_policy":{"","policy":{"null,"version":{"0"},"Writers":{"mod_policy":{"","
version":{"0"},"values":{"AnchorPeers":{"mod_policy":{"Admins","value":{"anchor_peers":{"host":"peer0.org2.example.com","por
t": 9051}]}]}],"version":{"0"},"values":{"{}","mod_policy":{"","policies":{"{}","values":{"{}","version":{"0"},"values":{"{}","mod_policy":{"","
","policies":{"{}","values":{"{}","version":{"0"},"mod_policy":{"","policies":{"{}","values":{"{}","version":{"0"},"values":{"{}","mod_policy":{"","

```


Step 9: Deploy Chaincode

```
./network.sh deployCC \  
-ccn basic \  
-ccp ../asset-transfer-basic/chaincode-javascript \  
-ccl javascript
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ cd $HOME/fabric-samples/test-network  
  
./network.sh deployCC \  
-ccn basic \  
-ccp ../asset-transfer-basic/chaincode-javascript \  
-ccl javascript  
Using docker and docker-compose  
deploying chaincode on channel 'mychannel'  
executing with the following  
- CHANNEL_NAME: mychannel  
- CC_NAME: basic  
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript  
- CC_SRC_LANGUAGE: javascript  
- CC_VERSION: 1.0  
- CC_SEQUENCE: auto  
- CC_END_POLICY: NA  
- CC_COLL_CONFIG: NA  
- CC_INIT_FCN: NA  
- DELAY: 3  
- MAX_RETRY: 5  
- VERBOSE: false  
executing with the following  
- CC_NAME: basic  
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript  
- CC_SRC_LANGUAGE: javascript  
- CC_VERSION: 1.0  
+ '[' false = true ']'  
+ peer lifecycle chaincode package basic.tar.gz --path ../asset-transfer-basic/chaincode-javascript --lang node --label basic_1.0  
+ res=0  
Chaincode is packaged  
Installing chaincode on peer0.org1...  
Using organization 1  
  
Checking the commit readiness of the chaincode definition successful on peer0.org2 on channel 'mychannel'  
Using organization 1  
Using organization 2  
+ peer lifecycle chaincode commit -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile /home/g2022_anuprita_mhapankar/fabric-samples/test-network/organizations/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem --channelID mychannel --name basic --peerAddresses localhost:7051 --tlsRootCertFiles /home/g2022_anuprita_mhapankar/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/tlsca/tlsca.org1.example.com-cert.pem --peerAddresses localhost:9051 --tlsRootCertFiles /home/g2022_anuprita_mhapankar/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem --version 1  
sequence 1  
+ res=0  
2026-04-09 15:02:00.641 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid [d578fad7d5ea9fbbab1c00b0f3bb78e4c57add2a11c8387b9bed0a90efd9559c] committed with status (VALID)  
localhost:7051  
2026-04-09 15:02:00.646 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid [d578fad7d5ea9fbbab1c00b0f3bb78e4c57add2a11c8387b9bed0a90efd9559c] committed with status (VALID)  
localhost:9051  
Chaincode definition committed on channel 'mychannel'  
Using organization 1  
Querying chaincode definition on peer0.org1 on channel 'mychannel'...  
Attempting to query committed status on peer0.org1, Retry after 3 seconds.  
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic  
+ res=0  
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':  
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]  
Query chaincode definition successful on peer0.org1 on channel 'mychannel'  
Using organization 2  
Querying chaincode definition on peer0.org2 on channel 'mychannel'...  
Attempting to query committed status on peer0.org2, Retry after 3 seconds.  
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic  
+ res=0  
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':  
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]  
Query chaincode definition successful on peer0.org2 on channel 'mychannel'  
Chaincode initialization is not required  
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 10: Set Peer Environment Variables

```
export  
FABRIC_CFG_PATH=$HOME/fabric-samples/config  
export PATH=$PATH:$HOME/fabric-samples/bin  
export CORE_PEER_TLS_ENABLED=true  
export CORE_PEER_LOCALMSPID="Org1MSP"  
export  
CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt
```

```
export
CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/organizations/
peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp
export CORE_PEER_ADDRESS=localhost:7051
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ export FABRIC_CFG_PATH=$HOME/fabric-samples/config
export PATH=$PATH:$HOME/fabric-samples/bin
export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"
export CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt
export CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp
export CORE_PEER_ADDRESS=localhost:7051
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 11: Initialize Ledger

```
peer chaincode invoke \
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile \
  $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles \
  $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \
  --peerAddresses localhost:9051 \
  --tlsRootCertFiles \
  $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \
  -c '{"function": "InitLedger", "Args": []}'
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \
--peerAddresses localhost:9051 \
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \
-c '{"function": "InitLedger", "Args": []}'
[INFO] 2024-08-08T09:04:00.000Z [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 12: Query All Assets

```
peer chaincode query -C mychannel -n basic -c '{"Args": ["GetAllAssets"]}'
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args": ["GetAllAssets"]}'
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"}, {"AppraisedValue":400,"Color":"red","ID":"asset2","Owner":"Brad","Size":5,"docType":"asset"}, {"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"asset"}, {"AppraisedValue":600,"Color":"yellow","ID":"asset4","Owner":"Max","Size":10,"docType":"asset"}, {"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"}, {"AppraisedValue":800,"Color":"white","ID":"asset6","Owner":"Michel","Size":15,"docType":"asset"}]
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 13: Read a Specific Asset

```
peer chaincode query -C mychannel -n basic -c
 '{"Args": ["ReadAsset", "asset5"]}'
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args": ["ReadAsset", "asset5"]}'
{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"}
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```


Step 14: Create a New Asset

```
peer chaincode invoke \  
  -o localhost:7050 \  
  --ordererTLSHostnameOverride orderer.example.com \  
  --tls \  
  --cafile  
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/exa  
mple.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cer  
t.pem \  
  -C mychannel \  
  -n basic \  
  --peerAddresses localhost:7051 \  
  --tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.e  
xample.com/peers/peer0.org1.example.com/tls/ca.crt \  
  --peerAddresses localhost:9051 \  
  --tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.e  
xample.com/peers/peer0.org2.example.com/tls/ca.crt \  
  -C  
'{"function": "CreateAsset", "Args": ["asset7", "blue", "10", "Anuprita", "500"]}'
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \  
-C mychannel \  
-n basic \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \  
-c '{"function": "CreateAsset", "Args": ["asset7", "blue", "10", "Anuprita", "500"]}'  
2024-04-04 15:09:15.874 UTC [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:"{'ID': 'asset7', 'Color': 'blue', 'Size': 10, 'Owner': 'Anuprita', 'AppraisedValue': 500}"  
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 15: Transfer Asset

```
peer chaincode invoke \  
  -o localhost:7050 \  
  --ordererTLSHostnameOverride orderer.example.com \  
  --tls \  
  --cafile  
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/exa  
mple.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cer  
t.pem \  
  -C mychannel \  
  -n basic \  
  --peerAddresses localhost:7051 \  
  --tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.e  
xample.com/peers/peer0.org1.example.com/tls/ca.crt \  
  --peerAddresses localhost:9051  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.e  
xample.com/peers/peer0.org2.example.com/tls/ca.crt
```

```
--peerAddresses localhost:9051 \
--tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.e
xample.com/peers/peer0.org2.example.com/tls/ca.crt \
-c '{"function":"TransferAsset","Args":["asset7","Sneha"]}'
```

```
q2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \
--peerAddresses localhost:9051 \
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \
-c '{"function":"TransferAsset","Args":["asset7","Sneha"]}'
2026-04-09 15:06:55.967 UTC [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:"Anuprita"
q2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 16: Stop the Network

```
cd $HOME/fabric-samples/test-network
./network.sh down
```

```
q2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ cd $HOME/fabric-samples/test-network
./network.sh down
Using docker and docker-compose
Stopping network.
project name can't be empty. Use '--project-name' to set a valid name
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing remaining containers
c72f6a9969f0
a811e27601bb
3f5198e286ca
3bcfb908afeb
8d2fc9e74375
Removing generated chaincode docker images
Untagged: dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbc664c87b24c035bee15cbcc7f59b629-c68f8ecc2a0ff0cc1497615f5e33a8bfa908e0fd3555f864cfa1398b52d341c
b:latest
Deleted: sha256:fa120af33c4db851b5dded871d445c39e7e2567eb3d753fcccfa850a32284087
Untagged: dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbc664c87b24c035bee15cbcc7f59b629-d3cf9b1f4b9747cc8433ac0e74bfaeade1cf0b97f5c1163835f604aa805d60d
c:latest
Deleted: sha256:edc7441a9a8c48b025dba4a79878631a900a26be317335ae9c9b2512c869f810
q2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

CONCLUSION:

The Hyperledger Fabric network was successfully implemented and chaincode was deployed. Asset operations such as creation, querying, and transfer were performed successfully. This experiment demonstrated secure and efficient asset management using a permissioned blockchain system.